

Pranav Upreti

437-326-1907 | pranav.upreti@mail.utoronto.ca | [linkedin.com/in/pranav-upreti](https://www.linkedin.com/in/pranav-upreti) | github.com/PRU1

EDUCATION

University of Toronto

Toronto, ON

Engineering Science, Engineering Physics (intended major) | 3.71 cGPA

2024 – 2029

- **Relevant Courses** : Process Modelling and Simulation (audited), ESC190 - Data Structures and Algorithms, CSC2332 - Introduction to Quantum Computing (auditing)
- **Skills**: MATLAB, C/C++, Python (Keras, TensorFlow, PennyLane, Qiskit, FastAPI, Streamlit), Java, SystemVerilog, RISC-V, Git, HTML/CSS/JS

EXPERIENCE

Embedded Programmer

November 2025 – Present

University of Toronto Biomedical Engineering Design Team (UTBIOME)

Toronto, ON

- Writing **C++** code for on demand droplet generator used in **medical research**.
- Implementing control logic for a centrifuge motor to control droplet generating through an **ESP32** microcontroller that interfaces with a centrifuge motor.

General Member

July 2025 – Present

University of Toronto Quantum Computing Club (UTQC)

Toronto, ON

- Contributed to a single qubit quantum maze solving algorithm based on a 2024 research for the Harvard Pacbots competition using **Python** and **Qiskit**
- Learned and implemented fundamental quantum algorithms like **grover's search** and **diffusion** during weekly meetings in PennyLane and Qiskit.

NRGHacks

Aug. 2023 – Nov. 2023

Event Organizer and Judge

Aurora, ON

- Co-founded an annual interschool hackathon with **100+ attendees**.
- Oversaw **~\$1000+** event budget and led a team of 7 to prepare workshops, handbooks, and event logistics.
- Ran an introductory workshop on Pygame to **~15** students.
- Judged participant entries on a panel with professional developers from industry.

PROJECTS

Virtual Aircraft Controller | *FastAPI, Streamlit, higgs-audio API*

- Developed a real-time, interactive virtual aircraft controller using **FastAPI** for backend and **Streamlit** for front end, complete with audio inputs and aircraft movement visualization
- Integrated speech to text models with NLP pipelines via **higgs-audio API** to extract heading, altitude, and runway assignment from speech input.
- Used **Git** for version control and collaboration, with detailed commit history to ensure project reproducibility.

The Second Brain | *HTML/CSS, L^AT_EX, React*

- Published and open sourced **400+** personal notes on programming, mathematics, quantum mechanics, etc. to share resources with peers. Built using HTML/CSS, **React**, and **Quartz** website.
- Published polished notes on **lagrangian mechanics** to accompany a lecture I delivered to **~15** Students. Built using **L^AT_EX**.

SlideHive | *CAD, Thermal Analysis with SimScale*

- A beehive redesigned for for the Toronto Beekeeping Collective to reduce lifting strain by **>50%** and maximize bee health
- Has similar thermal insulation to currently used Langstroth hives, tested using **thermal analysis** in SimScale
- Custom designed drawer mechanism that **maximizes longevity** by reducing propolis buildup (a resin bees produce).

Deets | *HuggingFace Pretrained LLMs*

- Developed an **AI-powered learning tool** to help students with special needs learn more effectively by summarizing content on their screen.
- Implemented a pretrained **BERT-CNN model** from HuggingFace to generate text summaries.
- Validated the idea through discussion with special needs educators in our school community.